Term 3 answer key

Formulas printed for students in every question. Answers with π are rounded to 1 decimal place, as each question says; probabilities are fractions in simplest form; every other answer is exact.

WEEK	SESSION	Q1 A Angles & shapes	Q2 B Measurement	Q3 C Number & percentages	Q4 D Algebra, ratio & rates	Q5 E Number plane, data & chance
1	1	45°	24 cm	12	$m = 18$	1/5
	2	70°	42 cm²	0.375	07:15	1/5
	3	38°	24 cm²	$17	32	(2, 5)
2	1	95°	96 cm³	85%	$150 and $90	4
	2	85°	120 cm²	0.09	$6x + 10$	23
	3	76°	58 cm²	$175	5 m/s	7/8
3	1	95°	25.1 cm	1000	75 beats per minute	6
	2	72°	40 cm	7/20	$m = 9$	9/10
	3	116°	750 mL	$16	11	1/3
4	1	115°	7 m	90%	2 : 3	2, 8
	2	124°	14 cm	$60	$6a + 6$	3
	3	130°	7 cm	−12	45 minutes	3/4
5	1	71°	5 cm	1240	$a = 7$	9
	2	53°	96 cm²	9%	60 and 12	20
	3	55°	78.5 cm	$6	34	9/25
6	1	150°	32 m	35%	25 seconds	1/2
	2	60°	121 m²	12	$7y + 21$	24
	3	66°	30 cm²	28 L	Box B ($23 per kg)	5/6
7	1	66°	2800 m	9/20	1 : 5	5, 7
	2	125°	91 m²	0.45	$x = 7$	23
	3	44°	70 cm²	$12	22	30
8	1	81°	100 m³	−11	8:50 pm	1/4
	2	72°	37.7 m	60%	$3x + 15$	1/5
	3	52°	28 cm	51 km	24 and 32	17/20
9	1	58°	56 cm²	35%	45 km/h	(3, 5)
	2	67°	400 m²	9 m	$x = 13$	8
	3	152°	48 cm²	81	6	9
10	1	118°	52 m²	370	$24 per hour	21/50
	2	105°	5 m²	26%	1 : 4	6
	3	125°	44.0 m	28 m	$4x − 12$	3/10

A MA4-ANG-C-01 complementary and supplementary angles, angles at a point, vertically opposite angles, angles on parallel lines; MA4-GEO-C-01 angle sum of a triangle and a quadrilateral, exterior angle of a triangle

B MA4-LEN-C-01 perimeter, circumference, length units; MA4-ARE-C-01 area of rectangles, squares, triangles, parallelograms and composite shapes, area units; MA4-VOL-C-01 volume of rectangular prisms, volume and capacity units

C MA4-INT-C-01 order of operations with integers; MA4-IND-C-01 powers, square and cube roots; MA4-FRC-C-01 fraction and percentage of a quantity, one quantity as a percentage of another, fractions, decimals and percentages, multiplying and dividing by 10, 100 and 1000*

D MA4-ALG-C-01 substitution, the distributive law; MA4-EQU-C-01 one- and two-step equations; MA4-RAT-C-01 simplifying ratios, sharing in a ratio, rates, best buys, speed; 24-hour time and time differences*

E MA4-LIN-C-01 tables of values, reading points on the number plane; MA4-DAT-C-01 dot plots; MA4-DAT-C-02 mean, median, mode and range; MA4-PRO-C-01 probability, complementary events, experimental probability*

Codes are from the NSW Mathematics K–10 Syllabus (2022). The NESA website could not be reached from the build machine, so **none of these codes was confirmed on the NESA site**; they match the codes used in the Year 9 and Year 10 keys.

* the topic is in this outcome as far as could be checked, but where it sits in the syllabus (or, for 24-hour time, whether it is Stage 3 revision with no Stage 4 code) could not be confirmed.